

Refinement Changelog

No.	Previous Version	Refinement / Improvement	Purpose / Impact
1	Learning system only used basic vocabulary lists with a single learning method	Redesigned the learning experience with roadmap-based flashcard progression and AI Voice Practice for real-time conversation training	Creates a more interactive, engaging, and realistic language learning experience
2	No AI assistant feature available	Added ENA AI Chatbot for real-time assistance related to language, culture, and app guidance	Enhances user support and provides personalized learning assistance
3	No fluency assessment system	Added Fluency Detector (Basic, Intermediate, Advanced) with certification support	Allows users to track progress and receive recognition based on proficiency level
4	No accessibility customization features	Added Disability Menu with accessibility support features for users from different backgrounds and needs	Improves inclusivity and ensures wider accessibility for all users
5	No communication feature between users	Added Direct Messaging feature for communication and social interaction	Encourages collaborative learning and community engagement
6	No user retention or motivation system	Added Streak feature with social sharing functionality	Encourages consistent learning habits and improves user retention
7	Application only supported a single language option	Added language selection feature with future expansion planning for additional languages	Supports scalability and allows broader user accessibility

8	No official learning recognition or external collaboration	Proposed collaboration with PUSTAKA to provide expert certification for users who complete required assessments	Increases platform credibility, user satisfaction, and official recognition of learning achievements
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